A Probabilistic Framework for Estimating the Modal Age at Death

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Abstract

The modal age at death is used to track longevity, but existing methods report it as a single number. We estimate its posterior distribution. Death counts are modeled as Poisson given the death rates and exposures, and posterior draws of the rates, passed through the life table, give the probability that each age is the mode and a credible interval for an interpolated modal age. With independent gamma priors, the posterior is computed exactly and concentrates on the true mode as the population grows. Because single-year rates are noisy, we also use a second-order random-walk prior on the log rates, which shares information across neighboring ages and estimates the degree of smoothing from the data; this posterior is computed with a Laplace approximation, which we check against posterior sampling. In simulations, the random-walk posterior mean was 1.4 to 10.6 times more accurate than the posterior mean under independent priors. When the distribution of deaths was unimodal, its errors were close to those of P-spline and Gompertz estimates, which were slightly more accurate; with a secondary peak, P-splines were the most accurate. The random-walk credible intervals were conservative, covering the true mode in 98% to 100% of replications, whereas bootstrap intervals for the other estimators either covered it less often or were much wider. For women in Denmark, France, Japan and Sweden since 1960, the random-walk estimates hardly changed with the starting age of the estimation, and the posterior gives the probability that the modal age declined in 2020.
Keywords: Modal age at death, Bayesian inference, Poisson–gamma model, random-walk prior, life table, uncertainty quantification

1 Introduction

The modal age at death, the age at which the life-table distribution of deaths peaks, has become a standard indicator of longevity. Unlike life expectancy, it does not depend on mortality at young ages and is determined by old-age mortality, so it follows the postponement of

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deaths into later life (Kannisto, 2001; Canudas-Romo, 2008; Horiuchi et al., 2013). Its trends have been used to separate the shifting of the distribution of deaths from its compression (Canudas-Romo, 2008; Bergeron-Boucher et al., 2015).

The mode is harder to estimate than a mean. The distribution of deaths is flat near its peak, so small random fluctuations in single-year death counts can move the highest point by a year or more. Three types of methods address this. Parametric models such as the Gompertz give the mode in closed form from the fitted parameters (Canudas-Romo, 2008; Missov et al., 2015). Smoothing methods fit a P-spline to the death counts and take the maximum of the smoothed distribution (Ouellette and Bourbeau, 2011; Horiuchi et al., 2013). A third group works with the life-table deaths directly: Kannisto (2001) interpolates between the ages around the highest count, and the discretized derivative tests (DDT) of Vazquez-Castillo et al. (2024) locate the age at which discrete versions of the conditions that define a mode are met.

These methods are usually presented as point estimators, with uncertainty assessed separately by the bootstrap or the delta method (Andreev and Shkolnikov, 2010; Keyfitz and Caswell, 2005). The resulting confidence intervals describe how much an estimate would vary over hypothetical repetitions of the data; they do not assign a probability to each candidate age. A single number also hides the common situation in which two or three neighboring ages are almost equally supported by the data.

We propose to estimate the probability distribution of the modal age at death. The mode is a function of the age-specific death rates, and the rates are uncertain because they are estimated from a finite number of deaths. Describing that uncertainty with a posterior distribution for the rates and carrying it through the life table gives, for each age, the probability that it is the modal age, together with a credible interval on a continuous age scale. The ingredients are established. Andreev et al. (2015) passed posterior distributions of death rates through the life table to obtain uncertainty limits for life expectancy, and Bayesian P-splines carry the uncertainty of a smoothed schedule and of its smoothing parameter into estimates (Lang and Brezger, 2004). We apply these ideas to the modal age, which, unlike life expectancy, jumps between ages when two values of the life-table deaths cross. We use two versions. In the first, the rates have independent gamma priors; the computation is then exact and some results can be derived in closed form. In the second, a random-walk prior on the log rates lets neighboring ages share information, in the same way as the penalties behind P-splines.

We compare the two versions with the Gompertz, P-spline and DDT estimators in a simulation study with a known modal age and in data for women in four countries. The comparison concerns methods for estimating the modal age and describing its uncertainty. Each method is judged by the accuracy of its estimate, by how often its interval contains the true mode and how wide that interval is, and by how much the estimate depends on the age range used. Section 2 develops the method, Section 3 reports the simulation study, Section 4 the application, and Section 5 discusses when the method should be used.

2 A Posterior Distribution for the Modal Age at Death

2.1 The modal age is a function of the death rates

Let $x = x_0, x_0 + 1, \ldots, \omega - 1$ index single-year age intervals $[x, x + 1)$, starting at an age x_0 and ending before the open-ended interval $\omega+$. For each age we observe the number of deaths D_x and the person-years of exposure E_x. Let m_x be the unknown death rate at age x, and write $\boldsymbol{m} = (m_{x_0}, \ldots, m_{\omega-1})^\top$, $\boldsymbol{D}$ and $\boldsymbol{E}$ for the corresponding vectors. There are $K = \omega - x_0$ ages.

Assuming a constant force of mortality within each interval, the life table is

$$q_x = 1 - e^{-m_x}, \qquad \ell_x = \exp\Big(- \sum_{y=x_0}^{x-1} m_y \Big), \qquad d_x = \ell_x \, q_x, \tag{1}$$

with $\ell_{x_0} = 1$ (Preston et al., 2001), so that d_x is the probability that a person alive at age x_0 dies in the interval $[x, x + 1)$. The modal age at death is the age with the largest d_x,

$$M(\boldsymbol{m}) = \operatorname*{arg\,max}_{x_0 \le x < \omega} \; d_x(\boldsymbol{m}). \tag{2}$$

Mortality below x_0 multiplies every d_x by the same factor, the probability of surviving from birth to x_0, and does not move the maximum. The starting age x_0 is therefore needed only to exclude the peak of deaths in infancy, and the mode depends on the rates at ages x_0 and above alone. The target is also the *life-table* distribution of deaths, not the observed distribution $D_x / \sum_y D_y$, which reflects the age structure of the population as well as its mortality.

If $\boldsymbol{m}$ were known, M would be a fixed number. What is uncertain is our knowledge of $\boldsymbol{m}$, which is estimated from a finite number of deaths. In a Bayesian analysis this uncertainty is described by a probability distribution for the rates given the observed data, the posterior distribution $p(\boldsymbol{m} \mid \boldsymbol{D})$ (exposures are treated as known and left implicit). Every rate schedule that is consistent with the data has its own modal age, and the probability that age x is the mode is the posterior probability of the schedules whose mode is x:

$$\mathbb{P}(M = x \mid \boldsymbol{D}) = \int \mathbb{I}\{M(\boldsymbol{m}) = x\} \, p(\boldsymbol{m} \mid \boldsymbol{D}) \, d\boldsymbol{m}, \tag{3}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. Equation (3) is the central object of the paper. The next subsections specify $p(\boldsymbol{m} \mid \boldsymbol{D})$ and show how to evaluate the integral.

2.2 Death counts and the posterior distribution of the rates

Following Brillinger (1986), we model the death count at each age as

$$D_x \mid m_x \sim \text{Poisson}(m_x E_x), \tag{4}$$

independently across ages. This is the standard likelihood behind the occurrence–exposure rate D_x / E_x, which is its maximum likelihood estimator. We do not condition on the total number of deaths: conditioning independent Poisson counts on their sum gives a multinomial

distribution whose cell probabilities are the expected shares of *observed* deaths (Bishop et al., 2007), which mix mortality with age structure and are not the life-table d_x.

For each rate we take a gamma prior, $m_x \sim \text{Gamma}(\alpha, \beta)$, independently across ages. The gamma family is conjugate to the Poisson likelihood (Gelman et al., 1995), so the posterior is

$$m_x \mid \boldsymbol{D} \sim \text{Gamma}(\alpha + D_x,\ \beta + E_x), \tag{5}$$

independently across ages, with shape $\alpha + D_x$ and rate $\beta + E_x$. We use $\alpha = 1/2$ and $\beta = 0$, which corresponds to the improper Jeffreys prior $p(m_x) \propto m_x^{-1/2}$ for a Poisson rate (Gelman et al., 1995); the posterior (5) is proper whenever $E_x > 0$, including ages with no deaths. The posterior mean of m_x is then $(D_x + 1/2)/E_x$, essentially the observed rate, and its coefficient of variation is $1/\sqrt{D_x + 1/2}$: about 3% with 1,000 deaths at that age and 1% with 10,000. The prior matters only at ages with very few deaths.

The rates are independent a posteriori, but the life-table deaths are not: through ℓ_x, each d_x depends on every rate below age x. Draws of $\boldsymbol{m}$ passed through (1) carry this dependence automatically.

2.3 From the posterior of the rates to the distribution of the mode

Because the posterior (5) is a product of known gamma distributions, (3) can be evaluated by Monte Carlo integration with exact draws. For $s = 1, \ldots, S$, draw $m_x^{(s)}$ from (5) at every age, compute $\boldsymbol{d}^{(s)}$ from (1), and record $M^{(s)} = M(\boldsymbol{m}^{(s)})$. Then

$$\widehat{\mathbb{P}}(M = x \mid \boldsymbol{D}) = \frac{1}{S} \sum_{s=1}^{S} \mathbb{I}\{M^{(s)} = x\}. \tag{6}$$

The data are held fixed and only the rates vary, so the simulation computes an integral; it does not create new data sets. The standard error of each estimated probability is at most $1/(2\sqrt{S})$, or 0.005 with $S = 10^4$ draws.

The output is a probability for each age. It can be summarized by the most probable age and its probability, or by a credible set: the smallest set of ages whose probabilities add up to at least 0.95. For point estimates and intervals on a continuous age scale, we take in each draw the vertex of the parabola through the life-table deaths of the modal interval $[x, x+1)$, where $x = M^{(s)}$, and of its two neighbors, each placed at the midpoint of its interval ($x - 1/2$, $x + 1/2$ and $x + 3/2$):

$$\tilde{M}^{(s)} = x + \frac{d_x^{(s)} - d_{x-1}^{(s)}}{\left(d_x^{(s)} - d_{x-1}^{(s)}\right) + \left(d_x^{(s)} - d_{x+1}^{(s)}\right)}, \tag{7}$$

which is the interpolation of Kannisto (2001) applied to each draw. The posterior mean, median and mode of $\tilde{M}$ are point estimates of the modal age, and its 2.5% and 97.5% quantiles give a 95% credible interval. Three quantities should be kept apart: the modal interval M in (2), the interpolated mode $\tilde{M}$, and the mode of an underlying continuous lifespan density. $\tilde{M}$ approximates the last. Applied to exact Gompertz schedules without noise, with modes at 90, 90.25, 90.5 and 90.75, the interpolation is off by -0.013 to -0.029 years (Appendix G), which is small relative to the uncertainty in national data.

2.4 Local modes

The structure of the life table also gives an exact result for *local* modes. Age x is a local mode if $d_{x-1} < d_x > d_{x+1}$. The factors common to the three ages cancel in (1), each comparison involves only two neighboring rates, and under (5) these rates are independent. The posterior probability that x is a local mode therefore reduces to a one-dimensional integral (Proposition 2 in Appendix A). For small rates, age x is a local mode when

$$m_x - m_{x-1} \gtrsim m_x^2 \gtrsim m_{x+1} - m_x,$$

the discrete form of the condition $\mu'(M) = \mu(M)^2$ that defines the mode of a continuous lifespan density (Canudas-Romo, 2008) and that underlies one of the DDT conditions. Summing the local-mode probabilities over ages gives the posterior expected number of local modes, a measure of how rough the single-year schedule is. The global mode has no comparable reduction, because d_x depends on all rates below x; this is why we evaluate (3) by Monte Carlo.

2.5 Concentration in large populations

With unlimited data, the method should identify the modal age with certainty.

Proposition 1. *Let $\boldsymbol{m}^0$ be the true rates, and suppose that $\boldsymbol{d}(\boldsymbol{m}^0)$ has a unique maximizer $x^\star$, with gap $\delta = d_{x^\star}(\boldsymbol{m}^0) - \max_{j \neq x^\star} d_j(\boldsymbol{m}^0) > 0$. Let the data follow (4) with $\boldsymbol{m} = \boldsymbol{m}^0$ and exposures $E_x = c_x N$, $c_x > 0$. Then, as $N \to \infty$,*

$$\mathbb{P}(M = x^\star \mid \boldsymbol{D}) \xrightarrow{p} 1.$$

The proof is in Appendix B. The speed of concentration depends on the gap δ relative to the posterior standard deviation of $d_{x^\star} - d_j$, which decreases roughly with the inverse square root of the number of deaths. Because the distribution of deaths is flat near its peak, δ is often small, and the posterior can remain spread over two or three ages even in large populations. This spread is information that a point estimate does not convey: the data do not distinguish between these ages.

2.6 Posterior and bootstrap probabilities

A common alternative is the bootstrap: replicate counts are drawn from the estimated rates, $D_x^* \sim \text{Poisson}(\hat{m}_x E_x)$, the mode is recomputed in each replicate, and the frequency of each age is reported (Appendix C). The two computations look alike, but the roles of data and rates are reversed. The bootstrap holds the rates at their estimates and varies the data, so its frequencies describe the sampling variability of an estimator. The posterior holds the data at what was observed and varies the rates, so it gives the probability that each age is the modal age, given the data, the Poisson model and the prior. With many deaths and a clear gap between the most likely age and its neighbors, both concentrate on the same age. They can differ when neighboring ages are nearly tied, which is common because the distribution of deaths is flat near its peak. Neither construction guarantees intervals with nominal coverage in finite samples.

Both kinds of interval can, however, be judged in the same way when the true mode is known: by how often they contain it and by how wide they are. Section 3 uses these criteria to compare the credible intervals of the posterior with bootstrap intervals for the other estimators.

2.7 Borrowing strength across ages: a random-walk prior

Under independent priors, each rate is estimated from the deaths at its own age only, and the random fluctuation of single-year counts passes into the posterior of the mode. Near the peak this fluctuation can create several local maxima, and the posterior then spreads over ages that differ only by noise. Demographers do not usually treat the rates at neighboring ages as unrelated: at adult ages the logarithm of the death rate increases almost linearly with age (Gompertz, 1825; Missov et al., 2015), and graduation and smoothing methods exploit this regularity. A prior that encodes it lets each rate borrow strength from its neighbors.

Let $\eta_x = \log m_x$. We replace the independent gamma priors with a second-order random walk (RW2) on the log rates,

$$\eta_x - 2\eta_{x-1} + \eta_{x-2} \sim \mathcal{N}(0, \lambda^{-1}), \qquad x = x_0 + 2, \ldots, \omega - 1, \tag{8}$$

independently across ages, and keep the Poisson likelihood (4). The joint prior density is $p(\boldsymbol{\eta} \mid \lambda) \propto \lambda^{(K-2)/2} \exp(-\frac{\lambda}{2} \boldsymbol{\eta}^\top \boldsymbol{P} \boldsymbol{\eta})$, where $\boldsymbol{P} = \boldsymbol{\Delta}^\top \boldsymbol{\Delta}$ and $\boldsymbol{\Delta}$ is the $(K-2) \times K$ matrix of second differences (Rue and Held, 2005). The prior says that the rate of increase of mortality changes little from one age to the next. It does not penalize the level or the slope of $\boldsymbol{\eta}$: every Gompertz schedule, $\eta_x = \log a + bx$, has zero penalty, and we give these two directions a flat prior. The posterior is then proper whenever deaths are observed at two or more ages. The precision λ sets how strongly the log rates are pulled toward a straight line. As $\lambda \to \infty$ the log rates are restricted to a straight line, a Gompertz schedule whose two parameters remain uncertain; as $\lambda \to 0$ each rate is informed by its own age alone (with a flat prior on $\log m_x$ rather than the Jeffreys prior). The prior therefore spans the range between a parametric model and the unsmoothed schedule, and the posterior weights the smoothing strengths according to the likelihood and the prior.

For fixed λ, the posterior mode of $\boldsymbol{\eta}$ maximizes

$$\ell(\boldsymbol{\eta}) - \frac{\lambda}{2} \sum_{x=x_0+2}^{\omega-1} (\eta_x - 2\eta_{x-1} + \eta_{x-2})^2 , \qquad \ell(\boldsymbol{\eta}) = \sum_x \left(D_x \eta_x - E_x e^{\eta_x}\right), \tag{9}$$

which is the graduation criterion of Whittaker (1923) written for Poisson counts. The same second-order penalty underlies P-splines (Eilers and Marx, 1996; Camarda, 2012), and the RW2 prior is its Bayesian form (Lang and Brezger, 2004; Rue and Held, 2005), so the central estimate of the RW2 posterior is a smoothed schedule of the kind demographers already use. The difference from the usual practice lies in what is done with it. Smoothing first and then reading the mode off one fitted schedule ignores the uncertainty of that schedule. Under the RW2 prior, each posterior draw is a different smooth schedule consistent with the data, and the mode is computed for each draw. The smoothing strength λ is estimated from the data and its uncertainty is carried into the draws as well. The distribution of the mode

thus reflects both the noise in the death counts and the uncertainty about how smooth the underlying schedule is.

Both versions estimate the same quantity, the modal age of the unknown rate schedule; they differ in what they assume about that schedule. The RW2 prior assumes that it is smooth in the sense of (8) and regularizes the estimated rates by sharing information across neighboring ages. The assumption has a cost where the true schedule bends sharply, for example where the age-at-death distributions of two subpopulations overlap; Section 3 examines this case.

Computation. The RW2 posterior has no closed form. We approximate it in two layers, in the spirit of the integrated nested Laplace approximation of Rue et al. (2009). For fixed λ, the posterior mode $\hat{\boldsymbol{\eta}}_\lambda$ is found by penalized Fisher scoring, and the posterior of $\boldsymbol{\eta}$ is approximated by $\mathcal{N}(\hat{\boldsymbol{\eta}}_\lambda, \boldsymbol{H}_\lambda^{-1})$, where $\boldsymbol{H}_\lambda$ is the negative Hessian of the log posterior at the mode (Tierney and Kadane, 1986). For λ, we place a flat prior on $\log_{10}\lambda$ over an equally spaced grid and weight each grid value by the Laplace approximation of its marginal likelihood. Each posterior draw takes a value of λ from these weights, then $\boldsymbol{\eta}$ from the corresponding Gaussian, and passes $\boldsymbol{m} = \exp(\boldsymbol{\eta})$ through the life table. Algorithm 1 summarizes both versions of the method, which differ only in how the rates are drawn. Appendix D gives the details and compares the approximation with posterior sampling by Markov chain Monte Carlo, in national data and in small simulated populations.

Algorithm 1: Posterior distribution of the modal age at death

Input: Deaths D_x and exposures E_x for $x_0 \le x < \omega$; number of draws S; for RW2, a grid $\lambda_1, \dots, \lambda_J$

Output: $\widehat{\mathbb{P}}(M = x \mid \boldsymbol{D})$ for each age x; draws $\tilde{M}^{(1)}, \dots, \tilde{M}^{(S)}$

if *RW2 prior* **then**
  for $j = 1, \dots, J$ **do**
    Find $\hat{\boldsymbol{\eta}}_{\lambda_j}$ and $\boldsymbol{H}_{\lambda_j}$ by iterating (S7); compute the weight $w_j \propto p(\boldsymbol{D} \mid \lambda_j)$ from (S8) ;
  end
end
for $s = 1, \dots, S$ **do**
  if *independent gamma priors* **then**
    Draw $m_x^{(s)} \sim \text{Gamma}(\alpha + D_x,\ \beta + E_x)$ for every age ;
  else
    Draw j with probabilities $w_1, \dots, w_J$, draw $\boldsymbol{\eta}^{(s)} \sim \mathcal{N}(\hat{\boldsymbol{\eta}}_{\lambda_j}, \boldsymbol{H}_{\lambda_j}^{-1})$ and set $\boldsymbol{m}^{(s)} = \exp(\boldsymbol{\eta}^{(s)})$;
  end
  Compute $\boldsymbol{d}^{(s)}$ from (1), $M^{(s)} = \arg\max_x d_x^{(s)}$ and $\tilde{M}^{(s)}$ from (7) ;
end
$\widehat{\mathbb{P}}(M = x \mid \boldsymbol{D}) \leftarrow S^{-1} \sum_s \mathbb{I}\{M^{(s)} = x\}$ for each age x ;

Because the RW2 prior links neighboring ages, the fitted schedule depends on the age range used: the rates near x_0 influence the fit at older ages. Under independent priors, x_0 affects the posterior of M only through the ages it excludes. Section 4 examines how much

this matters in practice. The exact local-mode result of Section 2.4 also does not carry over, because the rates are dependent a posteriori; the Monte Carlo estimate (6) does.

3 Simulation Study

Design. Ages at death follow a mixture of two Gompertz distributions with the same slope $b = 0.14$,

$$f(x) = w\, g_1(x) + (1-w)\, g_2(x), \qquad g_k(x) = a_k e^{bx} \exp\Big(-\frac{a_k}{b}\big(e^{bx}-1\big)\Big), \quad a_k = b\, e^{-bM_k}, \tag{10}$$

where g_1 and g_2 have modal ages $M_1 = 90$ and $M_2 = 65$. The mixture describes a population made of two groups with different levels of mortality. We consider four scenarios, $w = 1$, 0.8, 0.7 and 0.6, in which 0%, 20%, 30% and 40% of deaths occur in the earlier group. With $w < 1$ the density has a secondary peak between ages 66 and 68, whose height is 35%, 52% and 75% of the main peak. The true modal age is 90 in every scenario. For a radix N, exposures are the expected person-years $E_x = N \int_x^{x+1} S(t)\, dt$, with S the survival function of (10), and death counts are drawn as $D_x \sim \text{Poisson}(m_x E_x)$, with m_x the true central death rate, for ages $0, \ldots, 109$ and 110+. We use $N = 10^3, 10^4, 10^5$ and 10^6, from the annual deaths of one sex in a small region to those of a large country, and replicate each of the 16 combinations of scenario and population size 500 times. All methods use the starting age $x_0 = 40$.

Methods compared. We compute the posterior with independent gamma priors ("raw posterior") and the posterior with the RW2 prior, each with $S = 5{,}000$ draws, and summarize each by the posterior mean, median and mode of $\tilde{M}$ and the 95% credible interval. These are compared with three point estimators. For the Gompertz estimator, $\mu(x) = ae^{bx}$ is fitted by Poisson maximum likelihood to ages x_0 and above, and the mode is $\log(b/a)/b$ (Canudas-Romo, 2008; Missov et al., 2015). For the P-spline estimator, the death counts are smoothed with a Poisson P-spline, with the smoothing parameter chosen by the Bayesian information criterion as in Camarda (2012), and the mode is the maximum of the fitted lifespan density (Ouellette and Bourbeau, 2011). DDT is the method of Vazquez-Castillo et al. (2024), to which we add half a year. The published method treats d_x as the density of deaths at exact age x, whereas d_x counts the deaths between ages x and $x+1$ and is centered at $x + 1/2$; without the shift, DDT estimates are about half a year too low, and the other methods already place the deaths of $[x, x+1)$ at $x + 1/2$ (Appendix G). This shift and our rule for matching the zeros of the DDT conditions (Appendix G) make our implementation a modified version of the published method. For the three point estimators, 95% confidence intervals are obtained by a plug-in Poisson bootstrap: replicate counts $D_x^* \sim \text{Poisson}(D_x)$ with fixed exposures, 200 replicates, and the 2.5% and 97.5% quantiles of the re-estimated modes. For the P-spline, the smoothing parameter is held at the value selected on the original data. Accuracy is measured by the bias and the mean absolute error with respect to the true mode, and intervals by their coverage and mean width. With 500 replications, the Monte Carlo standard error of a coverage near 0.95 is about 0.01 (Morris et al., 2019).

Choice of posterior summary. For the RW2 posterior, the mean, median and mode of $\tilde{M}$ are about equally accurate in the unimodal scenario (mean absolute errors of 0.08, 0.07

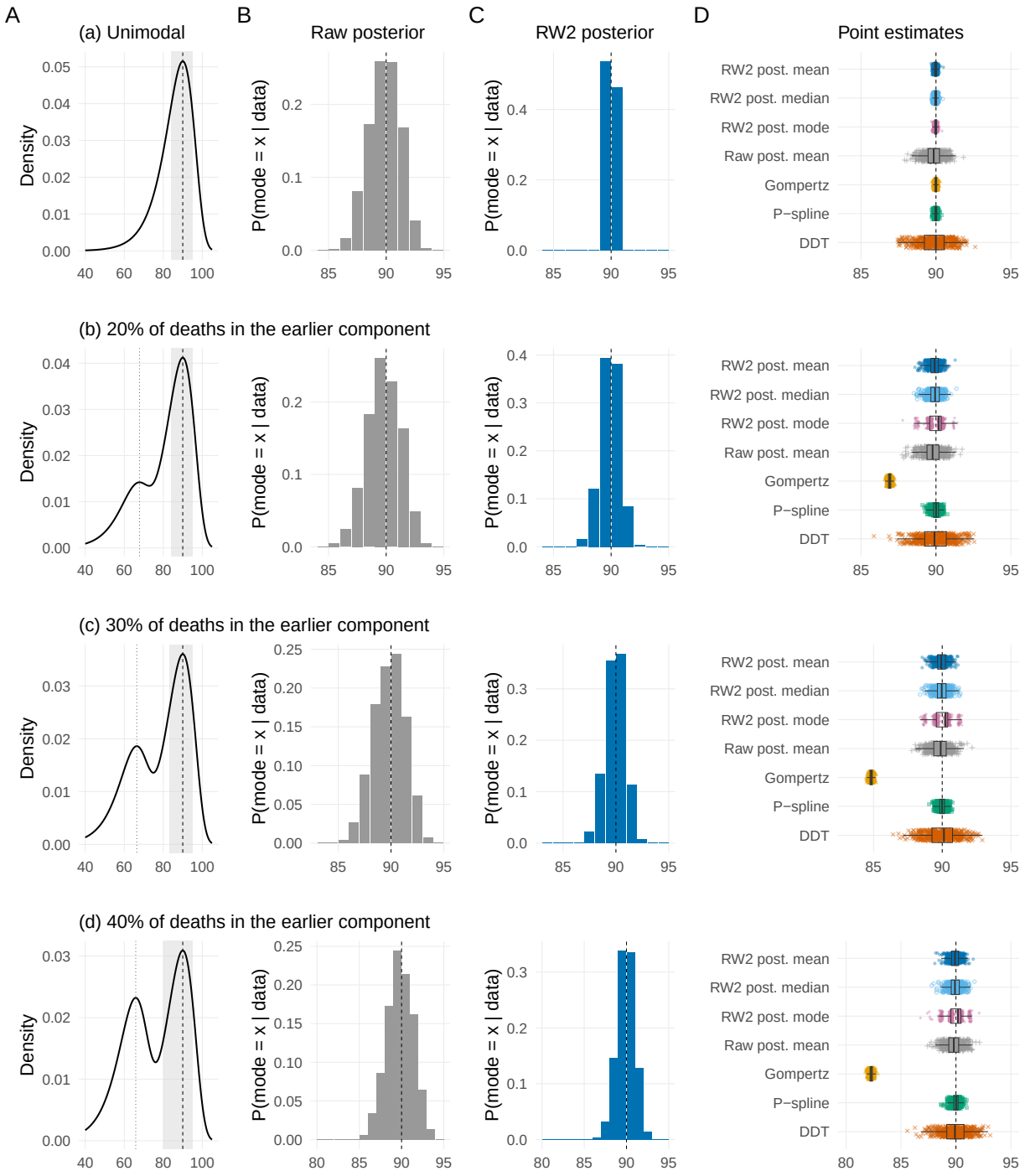


Figure 1: Simulation study with $N = 10^4$. Each row is a scenario. (A) True density of the age at death; the dashed line marks the true mode, the dotted line the secondary peak, and the shaded area the ages shown in B to D. (B) Posterior distribution of the modal age with independent gamma priors and (C) with the RW2 prior, averaged over 500 replications. (D) Point estimates across replications.

and 0.07 years at $N = 10^4$; Figure 2). With a secondary peak the posterior mean is the most accurate, with errors at $N = 10^4$ of 0.31, 0.35 and 0.39 years in scenarios (b) to (d), against

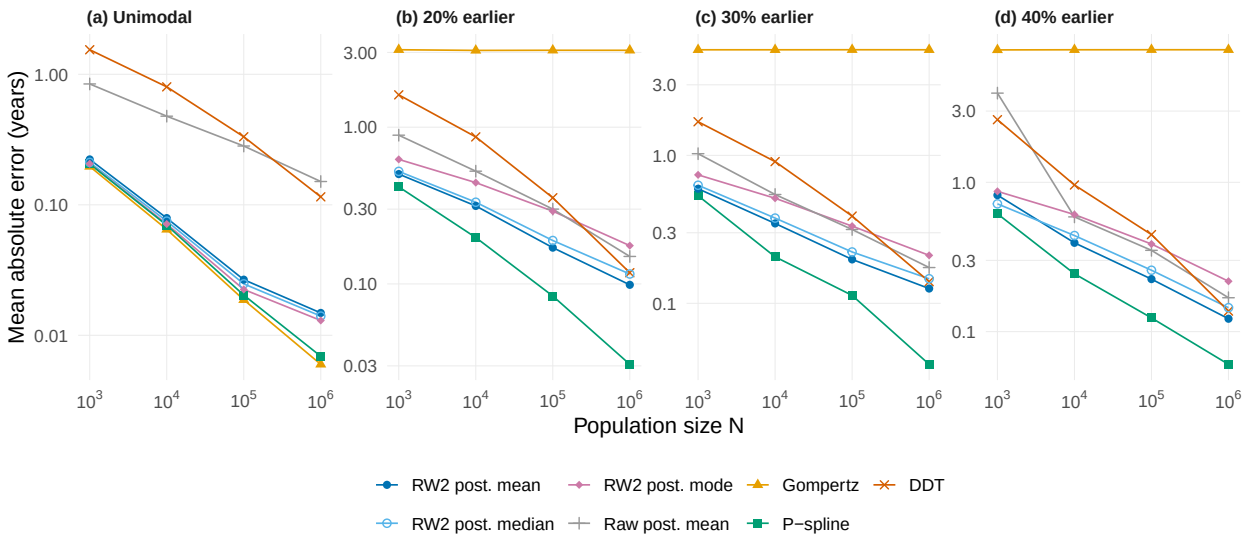


Figure 2: Mean absolute error of the estimated modal age by population size and scenario (logarithmic scales), 500 replications.

0.33 to 0.44 for the median and 0.44 to 0.61 for the mode. The only exception is scenario (d) with $N = 10^3$, where part of the posterior mass lies near the secondary peak and pulls the mean down (bias -0.35); the median is then more accurate (0.72 against 0.82). The raw posterior shows the same ordering. We therefore use the posterior mean as the point estimate.

RW2 against independent priors. Borrowing strength across ages concentrates the posterior of the mode (Figure 1, panels B and C). In the unimodal scenario, the mean absolute error of the RW2 posterior mean is 3.8 to 10.6 times smaller than that of the raw posterior mean, for example 0.08 against 0.48 years at $N = 10^4$. With a secondary peak the gain is a factor of 1.4 to 1.8, and much larger in scenario (d) with $N = 10^3$ (0.82 against 3.96 years). The raw posterior is biased downward in small populations (-0.48 years in the unimodal scenario at $N = 10^3$), whereas the bias of the RW2 posterior mean stays within 0.10 years of zero except in scenario (d) at $N = 10^3$.

Comparison of point estimates. When the distribution of deaths is unimodal, the RW2 posterior mean, the P-spline and the Gompertz estimators all have small errors, with the RW2 posterior mean slightly behind: mean absolute errors of 0.22, 0.21 and 0.20 years at $N = 10^3$, and 0.015, 0.007 and 0.006 years at $N = 10^6$. The Gompertz model is the true model in this scenario. In the largest populations, the remaining error of the RW2 posterior mean is mostly the bias of the interpolation (7) (-0.013 years at $N = 10^6$, against -0.0125 years for the noise-free schedule), which the P-spline and Gompertz estimators avoid by maximizing a continuous density. With a secondary peak the Gompertz model is misspecified, and its estimates are biased by -3.1, -5.2 and -7.7 years in scenarios (b) to (d), whatever the population size. After the half-year shift, DDT is nearly unbiased except at $N = 10^3$ (-0.26 years in the unimodal scenario and -1.18 in scenario (d)), but its estimates vary more from

one data set to the next: in the unimodal scenario its mean absolute error is 1.54, 0.80, 0.33 and 0.12 years from $N = 10^3$ to 10^6, seven to twelve times that of the RW2 posterior mean. With a secondary peak, the P-spline estimator is the most accurate: at $N = 10^4$ its mean absolute error is 0.20, 0.21 and 0.24 years in scenarios (b) to (d), against 0.31, 0.35 and 0.39 for the RW2 posterior mean. The difference is smaller at $N = 10^3$ (0.42 against 0.50 in scenario (b)).

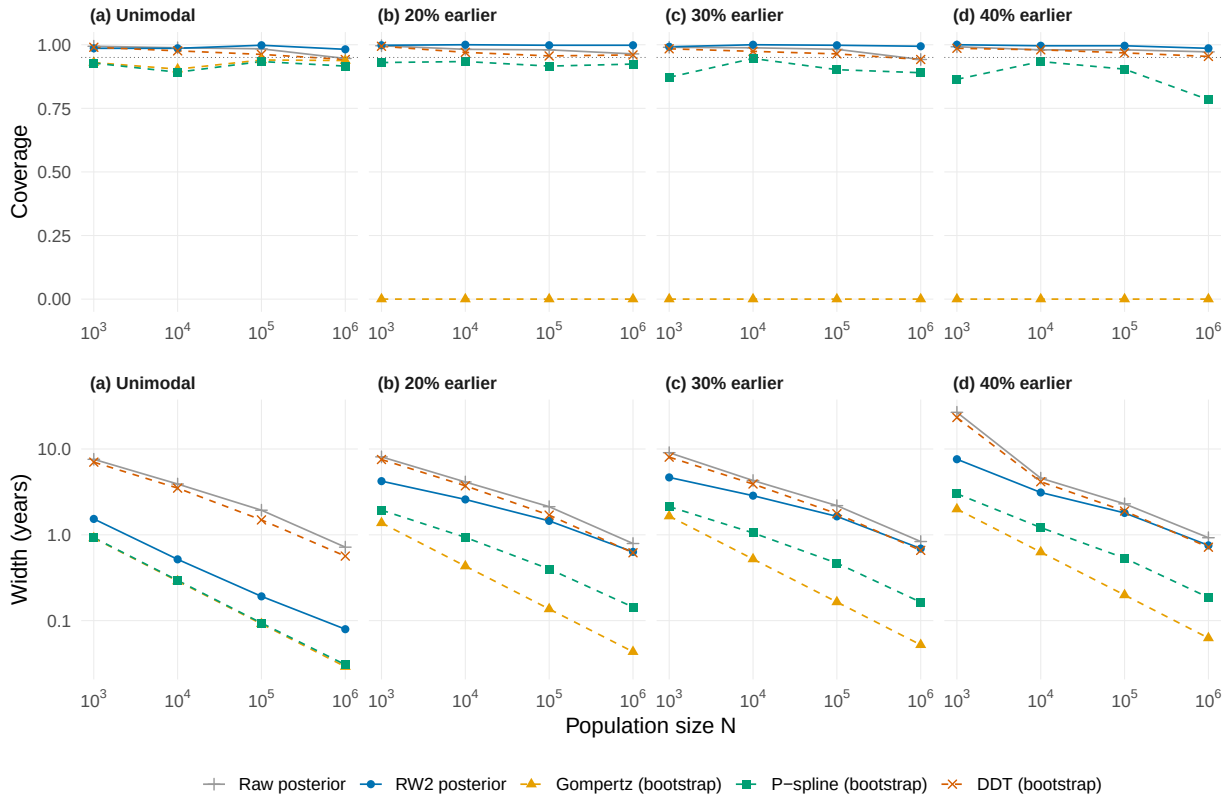


Figure 3: Coverage (top) and mean width (bottom, logarithmic scale) of 95% intervals for the modal age, by population size and scenario: credible intervals of the raw and RW2 posteriors (solid lines) and bootstrap confidence intervals of the Gompertz, P-spline and DDT estimators (dashed lines). The dotted line marks 95%.

Intervals. The coverage of the raw posterior credible intervals is close to the nominal level (0.94 to 1.00), and that of the RW2 intervals is conservative (0.98 to 1.00; Figure 3). The RW2 intervals are much narrower than the raw ones, 0.52 against 3.91 years in the unimodal scenario at $N = 10^4$. The bootstrap intervals of the other estimators behave differently. In the unimodal scenario, the Gompertz and P-spline intervals are narrower than the RW2 intervals (0.29 against 0.52 years at $N = 10^4$) and contain the true mode in 89% to 94% of replications, slightly below the nominal level. With a secondary peak, the Gompertz intervals never contain it: they are narrow (0.04 to 2.0 years) and centered on a biased estimate. The P-spline intervals then cover it in 78% to 95% of replications. Holding the smoothing parameter fixed in the bootstrap leaves out its uncertainty (Wood et al., 2016). In

a subset of settings, re-selecting it in every bootstrap sample raised the coverage to 0.96 and 0.97 in the unimodal scenario and to 0.94 in scenario (d) at $N = 10^3$, with intervals 1.6 to 2 times as wide, but not in scenario (d) at $N = 10^6$ (0.80 against 0.78; Appendix G). The DDT intervals reach nominal coverage (0.94 to 0.99) but are wide, 4.6 to 7.8 times wider than the RW2 intervals in the unimodal scenario (3.50 against 0.52 years at $N = 10^4$). Only the RW2 credible intervals stayed at or above the nominal level in every scenario and population size. The price is over-coverage with a secondary peak: the RW2 intervals (2.6 to 3.1 years wide at $N = 10^4$) are wider than the errors require, with coverage close to one and mean absolute errors of 0.31 to 0.39 years. One smoothing parameter has to fit both the bend between the two peaks and the main peak, which suggests that the marginal likelihood then favors weaker smoothing than the main peak alone would need.

4 Application

Data and design. We use deaths and exposures by single year of age ($0, \ldots, 109$ and 110+) from the Human Mortality Database (2026) for women in Denmark, France, Japan and Sweden, from 1960 to the latest year available (2025 for Denmark and 2024 for the other countries), 261 population-years in total. For each population-year we estimate the modal age with the RW2 posterior and with the Gompertz, P-spline and DDT estimators, each with the starting ages $x_0 = 40$, 50, 60 and 70. The modal age lies above all four, so the target is the same for every starting age (Section 2.1), and differences across starting ages measure how sensitive each estimator is to the age range it is given. DDT uses the rates at each age separately and does not depend on x_0; the other three fit a model across ages.

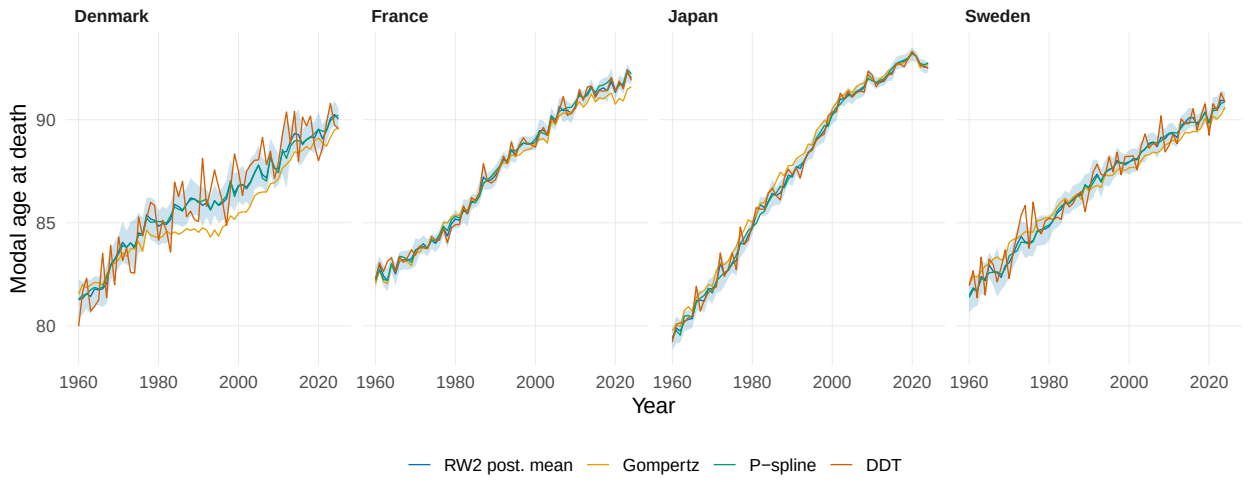


Figure 4: Modal age at death of women by country and year, starting age 40. The band is the 95% credible interval of the RW2 posterior.

Trends and agreement between methods. Between 1960 and the latest year, the RW2 posterior mean rose by 8.8 years in Denmark (81.3 to 90.0), 9.4 in Sweden (81.5 to 90.9), 9.7 in France (82.3 to 92.0) and 13.2 in Japan (79.4 to 92.5) (Figure 4). The median width of

the 95% credible interval is 1.1 years, ranging from 0.5 to 3.3 years; it is widest in Denmark (median 1.6 years), the smallest population, and narrowest in France and Japan (0.8 years). The P-spline estimate lies within the RW2 credible interval in 99.6% of population-years, with a mean absolute difference of 0.12 years from the RW2 posterior mean. The Gompertz estimate lies within it in 71.6% of population-years, on average 0.15 years lower. The DDT estimate lies within it in 81.2% of population-years, with a mean absolute difference of 0.39 years, and its series is visibly rougher from year to year (Figure 6), because DDT does not smooth the single-year rates. These percentages measure agreement between estimators applied to the same data, not accuracy.

Change between 2019 and 2020. The posterior also answers questions that a point estimate cannot, such as whether the modal age declined from one year to the next. Comparing the two marginal credible intervals is not enough for this: two intervals can overlap even when the difference is credibly different from zero (Schenker and Gentleman, 2001). Because the two years are fitted separately, we pair independent posterior draws of $\tilde{M}$ for 2019 and 2020 and compute the posterior of their difference. Table 1 and Figure 5 show the result. In France the modal age declined with posterior probability 0.99: the change was −0.52 years, with a 95% credible interval of −1.24 to −0.09, although the marginal intervals of the two years overlap. In Sweden the posterior mean fell by 0.54 years, and the probability of a decline was 0.94, with an interval for the change that includes zero (−1.25 to 0.16). In Denmark and Japan the posterior mean rose, and the probability of a decline was 0.22 and 0.19. The most probable single-year age was the same in both years in Denmark, France and Japan, and moved from 90 to 89 in Sweden.

Table 1: Modal age at death of women in 2019 and 2020 (RW2 posterior, starting age 40): most probable single-year age with its posterior probability, posterior mean of the change in $\tilde{M}$ from 2019 to 2020 with 95% credible interval, and posterior probability that the modal age declined.

	Most probable age (probability)		Change, 2019 to 2020	
Country	2019	2020	mean (95% interval)	P(decline)
Denmark	89 (0.6018)	89 (0.7065)	0.4471 (−0.7144, 1.5916)	0.2154
France	91 (0.8269)	91 (0.9508)	−0.5168 (−1.2422, −0.0932)	0.9932
Japan	93 (0.5567)	93 (0.8669)	0.2029 (−0.2509, 0.6294)	0.1851
Sweden	90 (0.8882)	89 (0.7317)	−0.5360 (−1.2509, 0.1599)	0.9353

Sensitivity to the starting age. Figure 6 shows the estimates for Denmark, the smallest of the four populations, and Table 2 summarizes all four (Figure S3 in the Appendix shows each country). The RW2 posterior mean and the P-spline estimator hardly change with the starting age. Moving it from 40 to 70 changes the RW2 posterior mean by 0.03 years on average and by at most 0.28 years, against 0.04 and 0.48 years for the P-spline estimator and 0.42 and 0.99 years for the Gompertz model. The Gompertz estimates also shift systematically, by +0.37 years on average, because the fitted slope depends on the age range; in Figure 6 this appears as four separate lines. Of the three posterior summaries, the posterior mode is the most sensitive (at most 0.57 years), in line with its lower accuracy in the simulation. For the RW2 posterior mean, the changes are more than an order of magnitude smaller than the

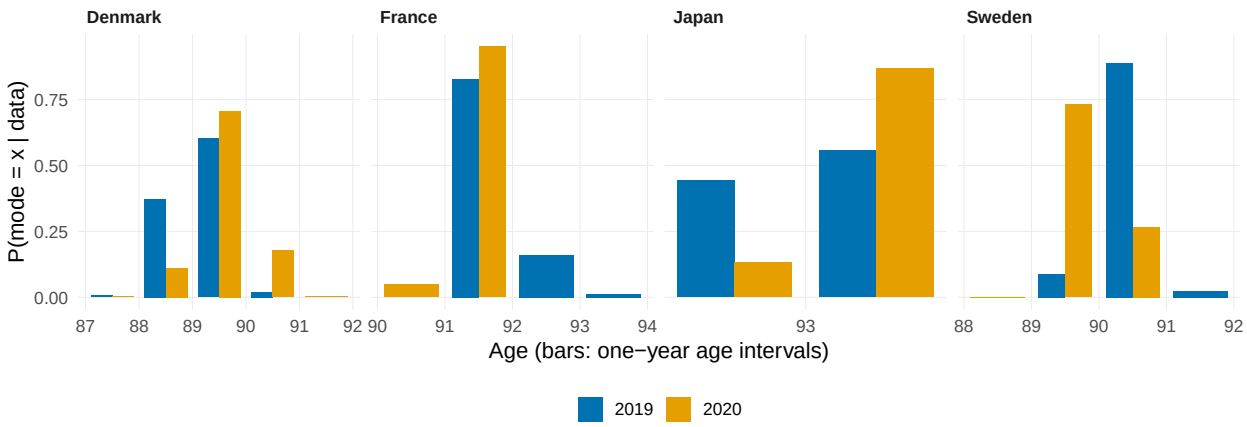


Figure 5: Posterior distribution of the modal age at death of women in 2019 and 2020 (RW2 prior, starting age 40): probability that each single-year age interval contains the mode.

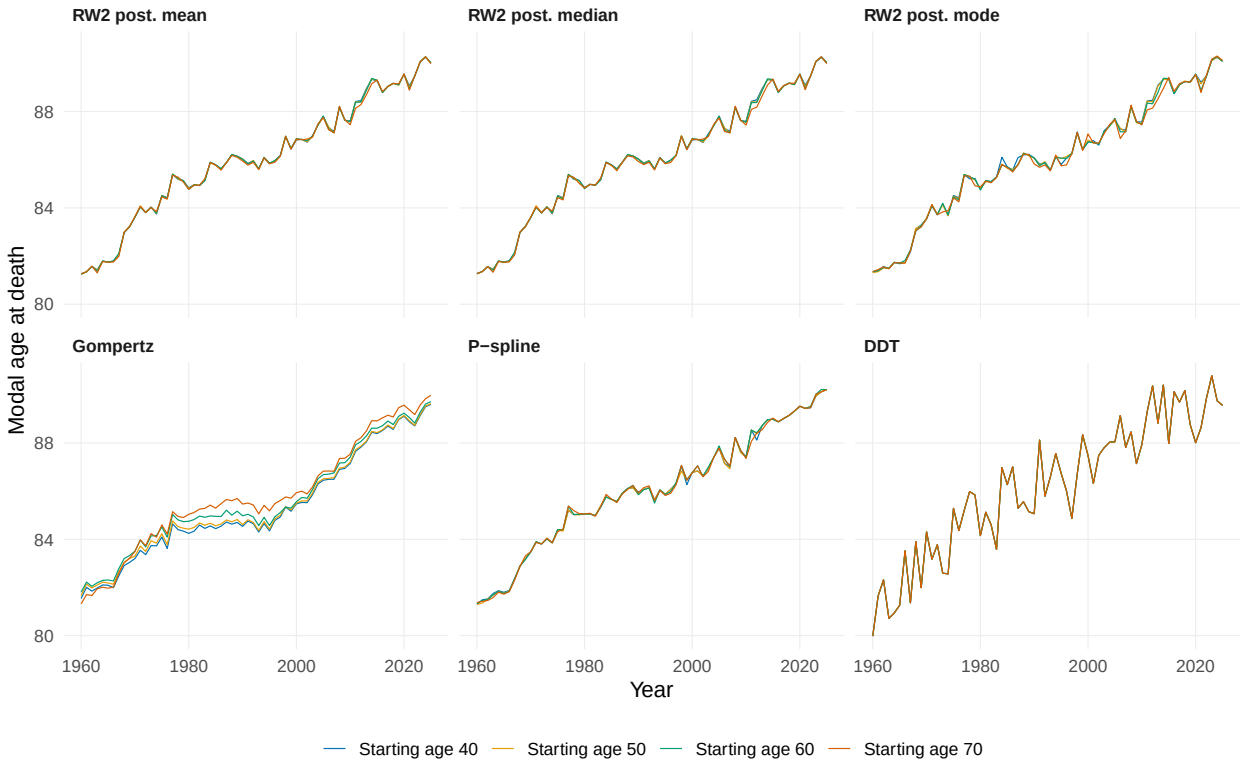


Figure 6: Modal age at death of women in Denmark by year, for each method (panels) and starting age (colors). Where the four lines coincide, the estimate does not depend on the starting age.

median width of the credible interval. This comparison measures stability, not accuracy: an estimator can be stable and biased, as the Gompertz model is with a secondary peak.

Table 2: Change in the estimated modal age at death when the starting age moves from 40 to 50, 60 and 70: mean absolute change and largest absolute change (years), women in 4 countries, all years (261 population-years). DDT uses the rates at each age separately and does not depend on the starting age.

	50 vs 40		60 vs 40		70 vs 40	
Method	Mean abs.	Max abs.	Mean abs.	Max abs.	Mean abs.	Max abs.
RW2 post. mean	0.0104	0.0931	0.0168	0.2273	0.0280	0.2825
RW2 post. median	0.0113	0.1026	0.0192	0.2496	0.0311	0.3396
RW2 post. mode	0.0263	0.3041	0.0425	0.4635	0.0521	0.5676
P-spline	0.0128	0.2623	0.0213	0.3043	0.0351	0.4754
Gompertz	0.1126	0.2421	0.2958	0.5860	0.4154	0.9948
DDT	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

5 Discussion

We proposed to estimate the modal age at death as a probability distribution rather than a single number. The distribution follows from two standard ingredients: a Poisson model for death counts given the rates, and the life table, which turns each rate schedule consistent with the data into a modal age. The result gives the probability of each age, a point estimate and a credible interval from the same computation.

The two priors make different assumptions about the same rate schedule. With independent gamma priors the posterior is exact, but the noise of single-year counts passes into it, which makes it useful mainly for showing how rough the schedule is near its peak. For estimating the modal age, the RW2 prior is the better choice. In the simulation study it reduced the mean absolute error of the posterior mean by a factor of 1.4 to 10.6 relative to independent priors and reduced the downward bias of the raw posterior in small populations. Among the posterior summaries, the posterior mean was the most accurate in almost all settings, and we recommend it as the point estimate.

Against existing estimators, the RW2 posterior mean had errors close to those of the P-spline and Gompertz estimators when the distribution of deaths was unimodal, slightly larger in the largest populations because of the interpolation. Unlike the Gompertz model, it does not restrict the log rates to a straight line: with a secondary peak the Gompertz estimates were biased by 3 to 8 years, while the bias of the RW2 posterior mean stayed within 0.10 years of zero except in the smallest populations. It is also less affected by single-year noise than the raw posterior and DDT. In data for women in four countries, it agreed closely with the P-spline estimates and was as stable as they were across starting ages, while the Gompertz estimates changed by 0.42 years on average. Its credible intervals were conservative, with coverage of 0.98 to 1.00 in every scenario and population size, whereas bootstrap intervals for the P-spline estimator with fixed smoothing covered the true mode in as few as 78% of replications (re-selecting the smoothing parameter improved this in most of the settings checked, at the cost of wider intervals), those of the Gompertz model never covered it when the distribution had a secondary peak, and those of DDT were four to eight times wider when it was unimodal.

The method is therefore most useful when the uncertainty of the modal age matters: comparing populations or years, judging whether a change from one year to the next exceeds the noise, as for the decline in France in 2020, or working with small populations, where

several ages can be almost equally supported by the data. In large populations with a unimodal distribution of deaths, its point estimate is close to that of a P-spline, and its contribution is the distribution and the interval that come with it.

The main limitation is the single smoothing parameter. When the distribution of deaths has a secondary peak, that parameter has to fit both the bend between the peaks and the main peak. The RW2 posterior mean was then less accurate than the P-spline estimator (0.31 to 0.39 against 0.20 to 0.24 years at $N = 10^4$), and its credible intervals were wider than needed. A smoothing precision that varies with age in (8) could improve this, but would need to be validated. The simulation used one family of schedules, mixtures of Gompertz distributions, so its results are a focused test rather than evidence for every mortality pattern. The intervals are also conditional on the exposures and the Poisson model; they do not include errors in the exposures or in reported ages. The same posterior draws give the distribution of any other quantity derived from the life table, such as life expectancy or lifespan variation, so the approach is not limited to the mode.

Acknowledgments

Funding: We gratefully acknowledge the financial support from the AXA Research Fund through the funding for the "*AXA Chair in Longevity Research*". This article is funded by the European Union (ERC, Born Once – Die Once, Grant agreement ID 101043983). Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

Author Contributions (CRediT): S.C.P.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing. P.V.C.: Conceptualization, Validation, Visualization, Software, Writing – review and editing.

Competing interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and materials availability: The code, simulation settings, and materials required to reproduce the analyses, tables, and figures are openly available at `https://github.com/scpatricio/modal_age_posterior`.

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Appendix

A Local modes and a Gaussian approximation

Age x is a local mode if $d_{x-1} < d_x > d_{x+1}$. Equation (1) gives $d_{x-1} = \ell_{x-1}(1 - e^{-m_{x-1}})$, $d_x = \ell_{x-1}e^{-m_{x-1}}(1 - e^{-m_x})$ and $d_{x+1} = \ell_{x-1}e^{-m_{x-1}}e^{-m_x}(1 - e^{-m_{x+1}})$. The common factors cancel, and each comparison involves only two neighboring rates:

$$d_x > d_{x-1} \iff m_{x-1} < u(m_x), \tag{S1}$$
$$d_x > d_{x+1} \iff m_{x+1} < v(m_x), \tag{S2}$$

where $u(m) = \log(2 - e^{-m})$, and $v(m) = -\log(2 - e^{m})$ for $m < \log 2$ and $v(m) = +\infty$ otherwise. Under (5), m_{x-1}, m_x and m_{x+1} are independent. Conditioning on m_x therefore makes the two events independent, and integrating over m_x gives the following result.

Proposition 2. *Let g_x and G_x denote the posterior density and distribution function of m_x in (5), with $G_x(+\infty) = 1$. Then*

$$\mathbb{P}(d_{x-1} < d_x > d_{x+1} \mid \boldsymbol{D}) = \int_0^\infty g_x(m)\, G_{x-1}\Big(u(m)\Big)\, G_{x+1}\Big(v(m)\Big)\, dm. \tag{S3}$$

The integral in (S3) is one-dimensional and can be evaluated to high accuracy with standard quadrature. For small rates, $u(m) \approx m - m^2$ and $v(m) \approx m + m^2$, which gives the condition for a local mode stated in Section 2.4.

Gaussian approximation. When many deaths are observed at each age, the posterior of $\boldsymbol{m}$ is approximately Gaussian with mean $\hat{\boldsymbol{m}}$, the vector of posterior means $(\alpha+D_x)/(\beta+E_x)$, and diagonal covariance $\boldsymbol{V}$ with entries $(\alpha + D_x)/(\beta + E_x)^2$. By the delta method, $\boldsymbol{d} \approx \mathcal{N}(\hat{\boldsymbol{d}}, \boldsymbol{\Sigma}_d)$ with $\hat{\boldsymbol{d}} = \boldsymbol{d}(\hat{\boldsymbol{m}})$ and $\boldsymbol{\Sigma}_d = \boldsymbol{J}\boldsymbol{V}\boldsymbol{J}^\top$. The Jacobian $\boldsymbol{J}$ is lower triangular:

$$\frac{\partial d_x}{\partial m_y} = \begin{cases} -d_x, & y < x, \\ \ell_{x+1}, & y = x, \\ 0, & y > x, \end{cases} \tag{S4}$$

evaluated at $\hat{\boldsymbol{m}}$. A higher rate at an earlier age lowers d_x by lowering survival to x; a higher rate at age x raises d_x; rates at later ages do not affect it.

Age x is the mode when all differences $d_x - d_j$, $j \neq x$, are positive. Let $\boldsymbol{\Lambda}_x$ be the $(K-1) \times K$ matrix whose rows are $\boldsymbol{e}_x - \boldsymbol{e}_j$ for $j \neq x$, where $\boldsymbol{e}_k$ is the k-th unit vector. Then $\boldsymbol{Y}_x = \boldsymbol{\Lambda}_x\boldsymbol{d} \approx \mathcal{N}(\boldsymbol{\Lambda}_x\hat{\boldsymbol{d}},\, \boldsymbol{\Lambda}_x\boldsymbol{\Sigma}_d\boldsymbol{\Lambda}_x^\top)$, and

$$\mathbb{P}(M = x \mid \boldsymbol{D}) \approx \int_{R_x} f_{\boldsymbol{Y}_x}(\boldsymbol{y})\, d\boldsymbol{y}, \tag{S5}$$

where $f_{\boldsymbol{Y}_x}$ is the Gaussian density of $\boldsymbol{Y}_x$ and $R_x = [0, \infty)^{K-1}$ is the positive orthant. Orthant probabilities of this kind can be computed with the algorithms of Genz and Bretz (2009), implemented in the `mvtnorm` package (Mi et al., 2009). The Genz–Bretz algorithm is itself a randomized quasi-Monte Carlo method, so (S5) does not remove simulation; its value

is interpretive. It shows that ages compete for the mode through the standardized gaps $(\hat{d}_x - \hat{d}_j)/\mathrm{sd}(d_x - d_j)$: two ages are both plausible modes when their life-table deaths differ by little relative to the posterior uncertainty of that difference. The approximation requires many deaths at the ages near the mode. When counts are small, (6) should be used.

B Proof of Proposition 1

Proof. Write $\bar{m}_x = (\alpha + D_x)/(\beta + c_x N)$ and $s_x^2 = (\alpha + D_x)/(\beta + c_x N)^2$ for the posterior mean and variance of m_x. Since $D_x/(c_x N) \to m_x^0$ in probability, $\bar{m}_x \to m_x^0$ and $s_x^2 \to 0$ in probability. Fix $\eta > 0$. On the event $|\bar{m}_x - m_x^0| < \eta/2$, Chebyshev's inequality gives $\mathbb{P}(|m_x - m_x^0| > \eta \mid \boldsymbol{D}) \le 4s_x^2/\eta^2$, and the probability of that event tends to one. A union bound over the K ages then gives $\mathbb{P}(\max_x |m_x - m_x^0| > \eta \mid \boldsymbol{D}) \to 0$ in probability.

The map $\boldsymbol{m} \mapsto \boldsymbol{d}(\boldsymbol{m})$ in (1) is continuous, so we can choose η such that $\max_x |m_x - m_x^0| \le \eta$ implies $|d_x(\boldsymbol{m}) - d_x(\boldsymbol{m}^0)| < \delta/2$ for all x. For such $\boldsymbol{m}$, $d_{x^\star}(\boldsymbol{m}) > d_{x^\star}(\boldsymbol{m}^0) - \delta/2 \ge d_j(\boldsymbol{m}^0) + \delta/2 > d_j(\boldsymbol{m})$ for every $j \ne x^\star$, so $M(\boldsymbol{m}) = x^\star$. Hence $\mathbb{P}(M = x^\star \mid \boldsymbol{D}) \ge 1 - \mathbb{P}(\max_x |m_x - m_x^0| > \eta \mid \boldsymbol{D}) \to 1$. □

C The bootstrap distribution of the mode

The parametric bootstrap of Section 2.6 reports the frequencies

$$\mathbb{P}^*(\hat{M}^* = x) = \sum_{\boldsymbol{D}^*} \mathbb{I}\{M(\boldsymbol{D}^*/\boldsymbol{E}) = x\}\, p(\boldsymbol{D}^* \mid \hat{\boldsymbol{m}}). \tag{S6}$$

where $p(\boldsymbol{D}^* \mid \hat{\boldsymbol{m}})$ is the Poisson distribution of the replicate counts given the estimated rates. Equations (3) and (S6) have a similar form, with the roles of data and rates reversed.

Both quantities are usually computed by simulation. The difference lies in what is averaged over: Equation (S6) averages over data sets that were not observed, holding the rates at their estimates, and Equation (3) averages over rates, holding the data at what was observed.

Sparse data make the difference concrete for the plug-in scheme used here, which resamples from the observed rates. If no deaths are observed at some age, then $\hat{m}_x = 0$ and every replicate also has zero deaths at that age, so this bootstrap treats the rate as known to be zero. The posterior (5) keeps a proper distribution, $\mathrm{Gamma}(1/2, E_x)$, which assigns positive probability to positive rates. A bootstrap from a fitted smooth schedule would not have this problem.

D Computation of the RW2 posterior

For fixed λ, the posterior mode $\hat{\boldsymbol{\eta}}_\lambda$ of Section 2.7 is found by penalized Fisher scoring, iterating

$$(\boldsymbol{W} + \lambda \boldsymbol{P})\, \boldsymbol{\eta}^{\mathrm{new}} = \boldsymbol{W}\boldsymbol{\eta} + \boldsymbol{D} - \boldsymbol{\mu}, \qquad \mu_x = E_x e^{\eta_x}, \quad \boldsymbol{W} = \mathrm{diag}(\boldsymbol{\mu}), \tag{S7}$$

until the largest change in $\boldsymbol{\eta}$ is below 10^{-7}. At the mode, $\boldsymbol{H}_\lambda = \boldsymbol{W}(\hat{\boldsymbol{\eta}}_\lambda) + \lambda \boldsymbol{P}$. The matrix $\boldsymbol{H}_\lambda$ is banded with bandwidth two, so each iteration, the Cholesky factor of $\boldsymbol{H}_\lambda$ and each draw

from $\mathcal{N}(\hat{\boldsymbol{\eta}}_\lambda, \boldsymbol{H}_\lambda^{-1})$ cost $O(K)$ operations. Each grid value of λ is weighted by the Laplace approximation of its marginal likelihood,

$$\log p(\boldsymbol{D} \mid \lambda) \approx \ell(\hat{\boldsymbol{\eta}}_\lambda) - \frac{\lambda}{2}\,\hat{\boldsymbol{\eta}}_\lambda^\top \boldsymbol{P}\hat{\boldsymbol{\eta}}_\lambda + \frac{K-2}{2}\log\lambda - \frac{1}{2}\log|\boldsymbol{H}_\lambda| + \text{const.} \tag{S8}$$

Comparison with posterior sampling. To check the approximation, we sampled the posterior of the same model (Poisson likelihood, RW2 prior, flat prior on the grid of $\log_{10}\lambda$) for women in Denmark, France, Japan and Sweden in 1960 and 2019, with starting age 50, and for four small simulated populations from Section 3 (scenarios (a) and (d), $N = 10^3$, two replications each, starting age 40). We used an independence Metropolis–Hastings sampler whose proposal is the Laplace approximation itself: a grid value λ_j is drawn with probability w_j, and $\boldsymbol{\eta}$ is drawn from $\mathcal{N}(\hat{\boldsymbol{\eta}}_{\lambda_j}, \boldsymbol{H}_{\lambda_j}^{-1})$. For the sampler, every grid value is evaluated and proposed with positive probability (w_j mixed with a uniform distribution with weight 0.01), so the chain targets the posterior on the full grid. The accept–reject step corrects the error of the approximation, and the acceptance rate measures how close the approximation is: it would be one if the approximation were exact. Each chain has 10^5 iterations.

In the eight national populations, the acceptance rates are between 0.93 and 0.98 (Table S1), and the effective sample size of $\tilde{M}$ is at least 83,000 of the 10^5 draws. The posterior means of $\tilde{M}$ from the two computations differ by at most 0.006 years and the limits of the 95% credible intervals by at most 0.03 years. The most probable age is the same in all eight cases, and its posterior probability differs by at most 0.005 (Figure S1); the posterior means of $\log_{10}\lambda$ differ by at most 0.002. In the four small simulated populations, the acceptance rates are lower, 0.88 to 0.97, and the posterior means differ by at most 0.05 years and the interval limits by at most 0.09 years, less than a tenth of the interval widths (1.3 to 4.5 years). The most probable age is again the same. The approximation is therefore accurate in national data and adequate in small populations, where its error is small relative to the posterior uncertainty.

E Full simulation results

Tables S2 to S5 report the mean absolute error, bias and standard deviation of every estimator, and the coverage and width of the credible and bootstrap intervals, for all scenarios and population sizes of Section 3. Figure S2 shows the bias by population size.

F Sensitivity to the starting age by country

Figure S3 extends Figure 6 to all four countries. It shows the estimates of each method by year at starting ages 40, 50, 60 and 70 and complements the summaries in Table 2.

G Implementation details

All computations are done in R. The code is available at the address given in the Data and materials availability statement.

Table S1: RW2 posterior of the modal age at death computed with the Laplace approximation and by posterior sampling (independence Metropolis–Hastings): posterior mean of $\tilde{M}$ with 95% credible interval, and acceptance rate of the sampler. Women, starting age 50; simulated populations (scenarios a and d of Section 3, $N = 10^3$), starting age 40.

		Modal age at death		Acceptance
Population	Year	Laplace	Sampling	rate
Denmark	1960	81.2632 (80.2467, 82.2879)	81.2572 (80.2150, 82.2818)	0.9659
Denmark	2019	89.1060 (88.3343, 89.9458)	89.1050 (88.3350, 89.9491)	0.9638
France	1960	82.3010 (81.7822, 82.9051)	82.3016 (81.7813, 82.9020)	0.9784
France	2019	91.8364 (91.5432, 92.4575)	91.8390 (91.5439, 92.4658)	0.9254
Japan	1960	79.3633 (78.8526, 79.8683)	79.3604 (78.8393, 79.8677)	0.9782
Japan	2019	93.0098 (92.7125, 93.2728)	93.0109 (92.7167, 93.2775)	0.9786
Sweden	1960	81.4744 (80.5841, 82.3046)	81.4767 (80.5745, 82.3039)	0.9686
Sweden	2019	90.3893 (89.8135, 91.0171)	90.3907 (89.8220, 91.0186)	0.9683
Simulated (a)	rep. 1	89.6449 (88.9667, 90.2585)	89.6517 (88.9688, 90.2483)	0.9698
Simulated (a)	rep. 2	89.7661 (89.1542, 90.6394)	89.7726 (89.1596, 90.6486)	0.9631
Simulated (d)	rep. 1	89.6827 (87.1238, 91.5788)	89.6948 (87.2130, 91.6044)	0.8878
Simulated (d)	rep. 2	89.2358 (87.4075, 91.2168)	89.2895 (87.4884, 91.2296)	0.8753

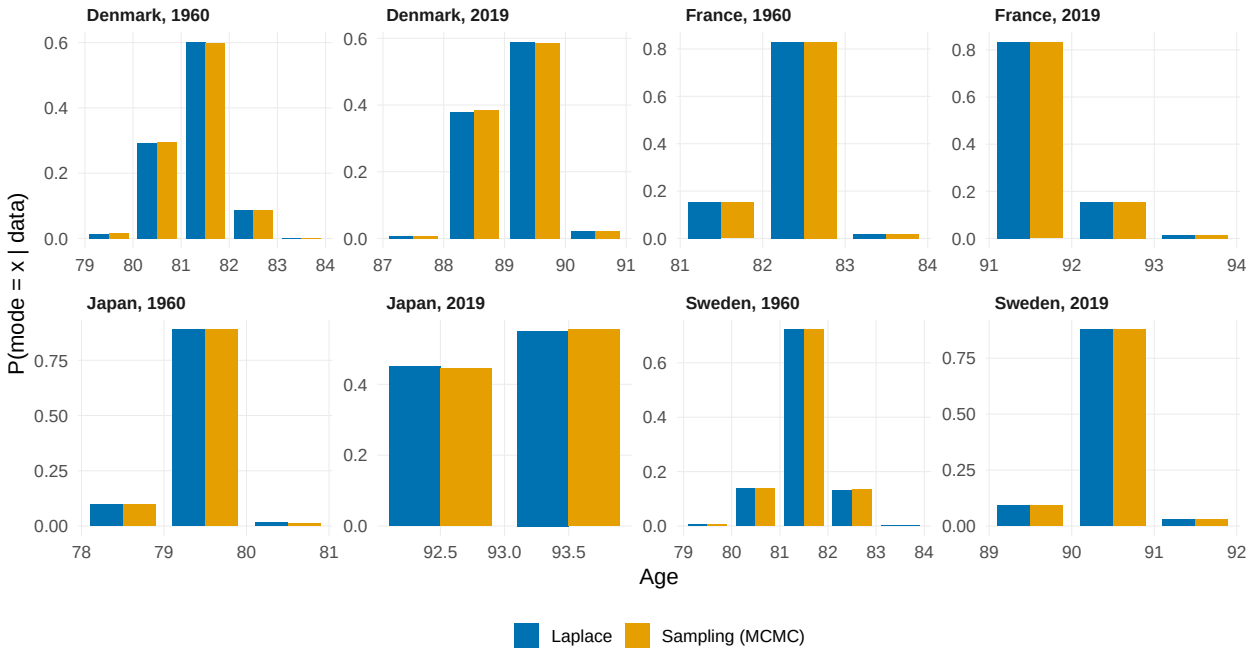


Figure S1: Posterior distribution of the modal age at death under the RW2 prior, computed with the Laplace approximation and by posterior sampling (independence Metropolis–Hastings), women, starting age 50.

Life table. All methods use the constant-force life table (1) on single years of age. The open interval 110+ has $q_x = 1$ and is never eligible as the mode. If the exposure is zero at some age above x_0, the schedule is truncated before that age, and the previous age becomes the open interval.

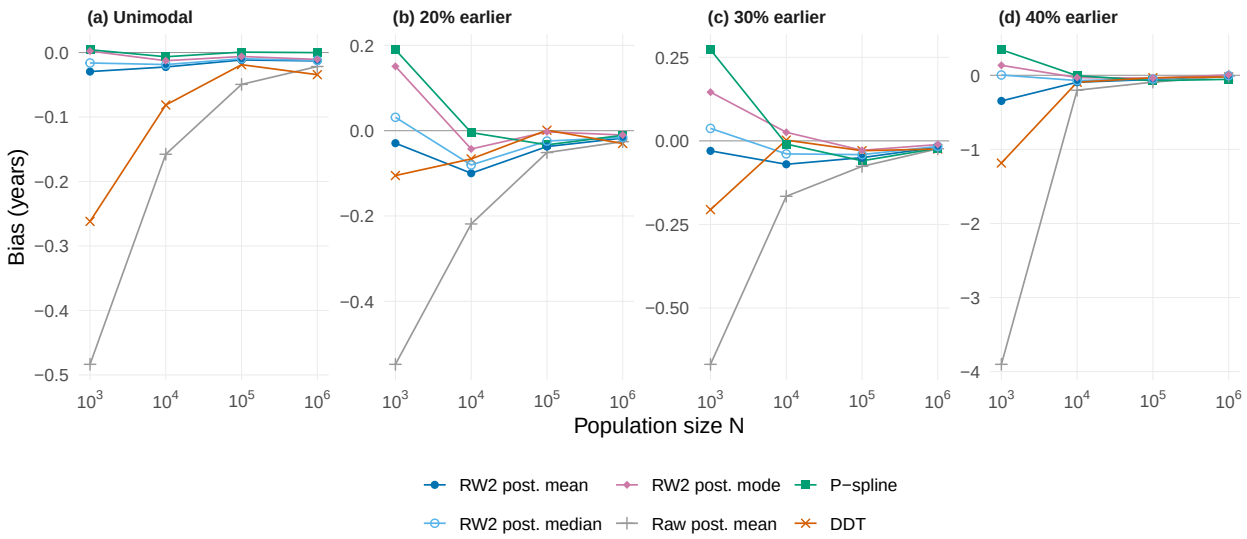


Figure S2: Bias of the estimated modal age by population size and scenario, starting age 40, 500 replications. Gompertz is not shown (its bias is 3 to 8 years with a secondary peak).

Table S2: Simulation study: mean absolute error (years) of the estimated modal age. Starting age 40, 500 replications, true mode 90.

		RW2 posterior			Raw posterior					
Scenario	N	Mean	Median	Mode	Mean	Median	Mode	Gompertz	P-spline	DDT
(a) unimodal	10^3	0.2227	0.2112	0.2065	0.8435	1.1589	1.6526	0.1967	0.2057	1.5423
	10^4	0.0791	0.0750	0.0713	0.4776	0.7004	0.9885	0.0648	0.0697	0.8015
	10^5	0.0267	0.0248	0.0225	0.2821	0.3828	0.5145	0.0187	0.0203	0.3321
	10^6	0.0148	0.0140	0.0130	0.1508	0.1794	0.2484	0.0060	0.0069	0.1151
(b) 20% earlier	10^3	0.5021	0.5226	0.6232	0.8893	1.1733	1.6887	3.1205	0.4162	1.6085
	10^4	0.3145	0.3322	0.4430	0.5239	0.7356	1.0204	3.0901	0.1977	0.8671
	10^5	0.1704	0.1892	0.2912	0.3001	0.4140	0.5682	3.0968	0.0837	0.3547
	10^6	0.0987	0.1157	0.1755	0.1497	0.1795	0.2563	3.0956	0.0307	0.1179
(c) 30% earlier	10^3	0.5949	0.6282	0.7385	1.0265	1.2772	1.7824	5.1728	0.5337	1.6878
	10^4	0.3467	0.3768	0.5131	0.5439	0.7710	1.0843	5.1736	0.2062	0.9078
	10^5	0.1980	0.2224	0.3325	0.3155	0.4430	0.5992	5.1765	0.1131	0.3887
	10^6	0.1259	0.1465	0.2109	0.1745	0.2060	0.2802	5.1748	0.0387	0.1401
(d) 40% earlier	10^3	0.8204	0.7172	0.8698	3.9555	1.9081	2.3054	7.7035	0.6152	2.6267
	10^4	0.3932	0.4400	0.6080	0.5866	0.8455	1.1326	7.7196	0.2439	0.9594
	10^5	0.2248	0.2577	0.3860	0.3502	0.4934	0.6506	7.7227	0.1236	0.4465
	10^6	0.1220	0.1446	0.2174	0.1686	0.2049	0.2851	7.7208	0.0604	0.1372

Posteriors. The raw posterior uses $\alpha = 1/2$ and $\beta = 0$. Both posteriors use $S = 5{,}000$ draws in the simulation study and $S = 10^4$ in the application. For the RW2 posterior, the grid of $\log_{10}\lambda$ runs from -1 to 8 in steps of 0.25; the search along the grid stops once the log marginal likelihood (S8) falls 12 units below its largest value, where the weights are below e^{-12}. The S draws are allocated to the grid values by multinomial sampling with the normalized weights. When the most probable age is the first or the last closed age, $\tilde{M}^{(s)} = x + 1/2$. The posterior mode of $\tilde{M}$ is the maximum of a Gaussian kernel density estimate of the draws with the default bandwidth of the R function `density`. The banded computations of Appendix D are coded in C++ through Rcpp. In the application (261 population-years, starting age 40), the posterior weight at either end of the grid was below 10^{-63}, grid values with weight above 0.001 lay between $\log_{10}\lambda = 1.5$ and 6.25, and the

Table S3: Simulation study: bias (years) of the estimated modal age. Starting age 40, 500 replications, true mode 90.

		RW2 posterior			Raw posterior					
Scenario	N	Mean	Median	Mode	Mean	Median	Mode	Gompertz	P-spline	DDT
(a) unimodal	10^3	−0.0295	−0.0162	0.0023	−0.4835	−0.4253	−0.4437	0.0104	0.0043	−0.2619
	10^4	−0.0224	−0.0186	−0.0127	−0.1577	−0.1372	−0.1399	−0.0051	−0.0065	−0.0813
	10^5	−0.0115	−0.0096	−0.0062	−0.0494	−0.0471	−0.0618	0.0025	0.0006	−0.0189
	10^6	−0.0130	−0.0122	−0.0107	−0.0217	−0.0183	−0.0147	0.0028	−0.0001	−0.0343
(b) 20% earlier	10^3	−0.0293	0.0313	0.1508	−0.5477	−0.3873	−0.3611	−3.1205	0.1897	−0.1052
	10^4	−0.0998	−0.0804	−0.0429	−0.2187	−0.1758	−0.1267	−3.0901	−0.0044	−0.0663
	10^5	−0.0373	−0.0243	−0.0027	−0.0513	−0.0354	−0.0404	−3.0968	−0.0332	0.0011
	10^6	−0.0175	−0.0159	−0.0101	−0.0257	−0.0230	−0.0189	−3.0956	−0.0109	−0.0301
(c) 30% earlier	10^3	−0.0299	0.0374	0.1458	−0.6686	−0.4828	−0.3930	−5.1728	0.2727	−0.2057
	10^4	−0.0700	−0.0390	0.0257	−0.1658	−0.0999	−0.0562	−5.1736	−0.0095	0.0022
	10^5	−0.0502	−0.0411	−0.0282	−0.0763	−0.0839	−0.0745	−5.1765	−0.0597	−0.0294
	10^6	−0.0189	−0.0166	−0.0110	−0.0238	−0.0174	−0.0089	−5.1748	−0.0221	−0.0244
(d) 40% earlier	10^3	−0.3460	0.0037	0.1347	−3.9016	−1.3594	−1.1522	−7.7035	0.3449	−1.1842
	10^4	−0.0935	−0.0709	−0.0308	−0.2014	−0.1687	−0.1807	−7.7196	−0.0051	−0.0941
	10^5	−0.0560	−0.0492	−0.0387	−0.0924	−0.0901	−0.1026	−7.7227	−0.0724	−0.0334
	10^6	−0.0100	−0.0042	0.0053	−0.0115	−0.0024	0.0095	−7.7208	−0.0545	−0.0217

Table S4: Simulation study: standard deviation (years) of the estimated modal age across replications. Starting age 40, 500 replications, true mode 90.

		RW2 posterior			Raw posterior					
Scenario	N	Mean	Median	Mode	Mean	Median	Mode	Gompertz	P-spline	DDT
(a) unimodal	10^3	0.2834	0.2703	0.2654	0.9491	1.3924	1.9961	0.2471	0.2601	1.9076
	10^4	0.0998	0.0949	0.0911	0.5872	0.8496	1.1721	0.0820	0.0891	0.9875
	10^5	0.0317	0.0298	0.0280	0.3530	0.4657	0.6000	0.0232	0.0258	0.4061
	10^6	0.0131	0.0126	0.0124	0.1820	0.2078	0.2625	0.0071	0.0095	0.1436
(b) 20% earlier	10^3	0.6359	0.6610	0.7460	0.9561	1.4220	2.0512	0.3304	0.4957	1.9893
	10^4	0.3885	0.4133	0.5195	0.6269	0.8742	1.2173	0.1139	0.2524	1.0709
	10^5	0.2241	0.2416	0.3218	0.3878	0.5121	0.6675	0.0362	0.1035	0.4495
	10^6	0.1206	0.1381	0.1891	0.1793	0.2079	0.2692	0.0117	0.0376	0.1489
(c) 30% earlier	10^3	0.7440	0.7758	0.8789	1.1161	1.5337	2.1615	0.4287	0.6081	2.0808
	10^4	0.4354	0.4730	0.6043	0.6734	0.9405	1.2985	0.1238	0.2631	1.1191
	10^5	0.2466	0.2713	0.3729	0.3841	0.5262	0.6990	0.0423	0.1262	0.4747
	10^6	0.1531	0.1733	0.2259	0.2111	0.2403	0.2978	0.0147	0.0430	0.1776
(d) 40% earlier	10^3	1.1451	0.8902	1.0500	2.8206	2.7760	4.1593	0.5202	0.6855	5.0485
	10^4	0.4954	0.5559	0.7426	0.7186	1.0339	1.3637	0.1689	0.3243	1.2007
	10^5	0.2815	0.3149	0.4482	0.4251	0.5907	0.7567	0.0510	0.1366	0.5526
	10^6	0.1504	0.1730	0.2316	0.2035	0.2370	0.3008	0.0170	0.0467	0.1759

posterior means of $\log_{10}\lambda$ ranged from 1.9 to 4.9. In the simulation, the posterior means ranged from 2.5 to 7.6; values near the upper end occur in the unimodal scenario, where the true log rates lie on a straight line.

Interpolation without noise. To measure the error of the interpolation (7) itself, we applied the life table (1) and (7) to the exact central death rates of single Gompertz distributions with $b = 0.14$ and modes at 90, 90.25, 90.5 and 90.75 (starting age 40). The interpolated modes are 89.987, 90.227, 90.471 and 90.724, errors of -0.013 to -0.029 years. Using the exact interval probabilities of the density instead of the constant-force life table changes these values by less than 0.001, so the error comes from the interpolation, not from the life table.

Table S5: Simulation study: coverage and mean width (years) of the 95% intervals for the modal age: credible intervals of the raw and RW2 posteriors (posterior of $\tilde{M}$) and bootstrap percentile intervals of the Gompertz, P-spline and DDT estimators. Starting age 40, 500 replications, true mode 90.

		Coverage					Width				
Scenario	N	Raw	RW2	Gomp.	P-spl.	DDT	Raw	RW2	Gomp.	P-spl.	DDT
(a) unimodal	10^3	0.9940	0.9860	0.9300	0.9280	0.9900	7.5840	1.5317	0.9174	0.9275	7.0263
	10^4	0.9880	0.9860	0.9040	0.8920	0.9760	3.9071	0.5177	0.2889	0.2941	3.4997
	10^5	0.9840	0.9980	0.9400	0.9340	0.9620	1.9354	0.1917	0.0917	0.0937	1.4913
	10^6	0.9460	0.9820	0.9380	0.9160	0.9420	0.7180	0.0795	0.0291	0.0307	0.5649
(b) 20% earlier	10^3	0.9960	0.9980	0.0000	0.9300	0.9940	8.0973	4.2211	1.3758	1.9391	7.5429
	10^4	0.9820	1.0000	0.0000	0.9340	0.9700	4.1291	2.5832	0.4316	0.9343	3.7257
	10^5	0.9800	0.9980	0.0000	0.9160	0.9560	2.1268	1.4527	0.1365	0.3995	1.6955
	10^6	0.9640	0.9980	0.0000	0.9240	0.9600	0.7918	0.6327	0.0434	0.1438	0.6178
(c) 30% earlier	10^3	0.9900	0.9920	0.0000	0.8720	0.9840	9.0573	4.6652	1.6504	2.1307	8.0380
	10^4	0.9880	1.0000	0.0000	0.9460	0.9740	4.2937	2.8579	0.5215	1.0526	3.9101
	10^5	0.9820	0.9980	0.0000	0.9020	0.9640	2.1876	1.6395	0.1647	0.4677	1.7615
	10^6	0.9420	0.9940	0.0000	0.8900	0.9420	0.8347	0.6911	0.0522	0.1639	0.6586
(d) 40% earlier	10^3	0.9920	1.0000	0.0000	0.8640	0.9860	26.7608	7.5979	1.9931	3.0086	23.2659
	10^4	0.9800	0.9960	0.0000	0.9340	0.9800	4.5709	3.1150	0.6284	1.2145	4.1567
	10^5	0.9800	0.9960	0.0000	0.9040	0.9680	2.2963	1.7993	0.1988	0.5348	1.8921
	10^6	0.9720	0.9860	0.0000	0.7840	0.9540	0.9247	0.7527	0.0627	0.1861	0.7169

Gompertz. The model $\log m_x = \log a + b(x+1/2)$ is fitted by Poisson maximum likelihood, with offset $\log E_x$, to the ages from x_0 to $\omega - 1$. The mode is $\log(b/a)/b$.

P-spline. The R package MortalitySmooth (Camarda, 2012) is no longer distributed through CRAN, so we coded the same model. The log rate at $x + 1/2$ is a combination of cubic B-splines on equally spaced knots, with $\lfloor(\omega - x_0)/5\rfloor$ intervals (at least four), and a second-order difference penalty on the coefficients. The model is fitted to the counts by penalized iteratively reweighted least squares with offset $\log E_x$, and the smoothing parameter minimizes the Bayesian information criterion over $\log_{10}\lambda \in \{-2, -1.5, \ldots, 7\}$. The mode is the maximum of the fitted density $f(t) = \mu(t)\exp\{-\int_{x_0}^{t}\mu(u)\,du\}$ on a grid of 0.01 years, refined by the vertex of the parabola through the three grid points around the maximum.

DDT. We compute the four conditions of Vazquez-Castillo et al. (2024, Equation 9) from the observed rates and the corresponding life-table deaths, and locate each change of sign from positive to negative by linear interpolation between integer ages. Every zero of the first-difference condition at which the second-difference condition is negative is a candidate local maximum. The zeros of the two rate conditions closest to the candidate, within two years, are averaged with it; a condition with no zero within two years is left out. Among the candidates, the estimate is the one with the largest life-table deaths. The original description does not specify how zeros are matched when the conditions cross zero more than once, so this matching rule is our choice. Finally, we add 1/2 to the estimate. The published conditions treat d_x as the density of deaths at exact age x, but d_x counts the deaths in $[x, x+1)$, whose center is $x + 1/2$; the Gompertz and P-spline fits evaluate the hazard at $x + 1/2$, and the interpolation (7) locates the mode within $[x, x + 1)$. Without the shift, DDT estimates were

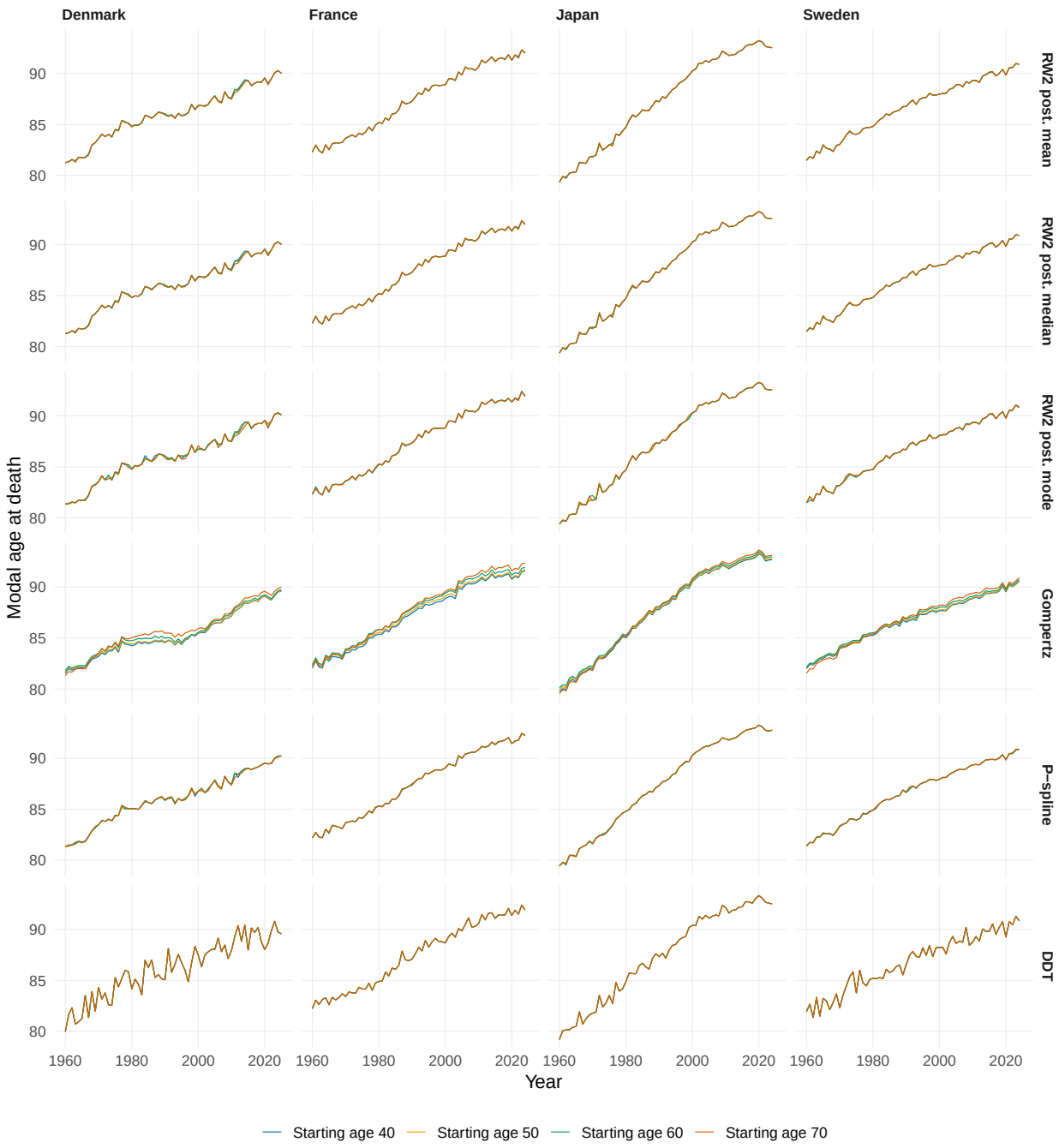


Figure S3: Modal age at death of women by year, method (rows) and country (columns), at starting ages 40, 50, 60 and 70.

on average about half a year below the true mode in the simulation study and below the other estimates in the application.

Bootstrap intervals. For the Gompertz, P-spline and DDT estimators, each simulated data set is resampled 200 times as $D_x^* \sim \text{Poisson}(D_x)$ with the exposures held fixed, the mode

is re-estimated in each replicate, and the 2.5% and 97.5% quantiles of the replicates form the 95% interval. With 200 replicates, each tail quantile rests on about five draws. For the P-spline, the smoothing parameter is fixed at the value selected on the original data. To see how much this matters, we repeated the P-spline bootstrap with 500 replicates, once with the smoothing parameter fixed and once re-selected by the Bayesian information criterion in every replicate, for the first 200 data sets of scenarios (a) and (d) at $N = 10^3$ and 10^6. With the smoothing parameter fixed, coverage was 0.94 and 0.91 in scenario (a) at $N = 10^3$ and 10^6, and 0.89 and 0.78 in scenario (d). With re-selection it was 0.96, 0.97, 0.94 and 0.80, and the mean width rose from 0.94 to 1.84 years and from 0.03 to 0.06 years in scenario (a), and from 3.27 to 5.28 years in scenario (d) at $N = 10^3$, but stayed at 0.19 years in scenario (d) at $N = 10^6$. The Monte Carlo standard errors of these coverages are 0.01 to 0.03.